- $v = \frac{dx}{dt}$, $a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$

$v = \int a\,dt = v_0 + \int_{t_1}^{t_2} a\,dt$

$x = \int v\,dt = x_0 + \int_{t_1}^{t_2} v\,dt$

$F = ma = m\frac{dv}{dt} = m\frac{d^2x}{dt^2}$

由此即可构造常微分方程

- $F = -kx$ 回复力

$F = ma = m\frac{d^2x}{dt^2} = -kx$

$\Rightarrow \frac{d^2x}{dt^2} + \frac{k}{m}x = 0$

$x = A_1\cos wt + A_2\sin wt$ or $x = A\cos(wt+\varphi)$

$w = \sqrt{k/m}$

$w = 2\pi f$, $T = 2\pi/w$

单摆: $w = \sqrt{g/L}$, $T = 2\pi\sqrt{L/g}$

- $dJ = F\,dt = ma\,dt = m\,dv$

$\rightarrow J = mv_2 - mv_1 = p_2 - p_1$

$p = mv$

$dJ = dp = F\,dt \rightarrow F = \frac{dp}{dt} = \frac{d(mv)}{dt} = ma + v\frac{dm}{dt}$

$W = \int F\,dx = \int \frac{dp}{dt}\cdot v\,dt = \int v\cdot dp = m\int v\cdot dv = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$

$K = \frac{1}{2}mv^2 = \frac{p^2}{2m}$

- 势能 $\Delta U = \underline{U_b - U_a = -W_{ab}} = \int_a^b -F\cdot dr$, $U = U(r)$

似是能量守恒

保守力: $\oint F\,dr = 0$

Gradient (梯度): $dU = -F\,dr = -(F_x\,dx + F_y\,dy + F_z\,dz)$

$F = F(r) = -\left(\frac{\partial U}{\partial x}i + \frac{\partial U}{\partial y}j + \frac{\partial U}{\partial z}k\right)$, $F = -\nabla U$

梯度算子: $\nabla = i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}$

- 引力: $F = -\frac{GMm}{r^2}\hat{r}$, $g = \frac{GM}{r^2}$

$U = -\frac{GMm}{r}$

势图:

平衡态: $F = -\frac{dU}{dx} = 0$

stability: $\frac{d^2U}{dx^2} \begin{cases} >0 & \text{stable} \\ <0 & \text{unstable} \\ =0 & \text{neutral} \end{cases}$

- $w = d\theta/dt$, $\alpha = dw/dt$

- 力矩 $\tau = r \times F$

转惯 $F_i = m_i a_i = m_i(\alpha \times r_i + a_{ci})$, $r_i \times F_i = m_i r_i \times (\alpha \times r_i) = m_i r_i^2 \alpha$

$\Sigma \tau = I\alpha$, $I = \Sigma_i m_i r_i^2$

$I = \int r^2 dm$, $dm = \begin{cases} \lambda dl \\ \sigma dA \\ \rho dV \end{cases}$

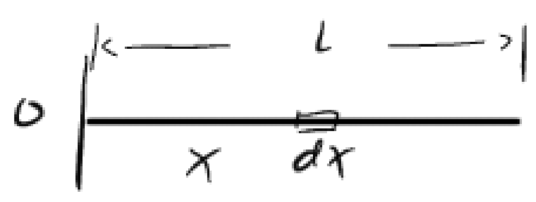
$K = \Sigma_i \frac{1}{2} m_i v_i^2 = \frac{1}{2}\Sigma_i m_i r_i^2 w^2 = \frac{1}{2} I w^2$

$dK = dW = F \cdot dl = F_t R\, d\theta = \tau \cdot d\theta$

(1) $\tau = 0$, w不变

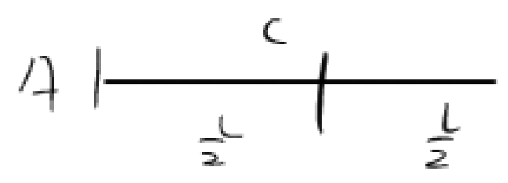
(2) $\alpha \propto \frac{\tau}{I}$

(3) $\tau = I\alpha = I\frac{dw}{dt}$



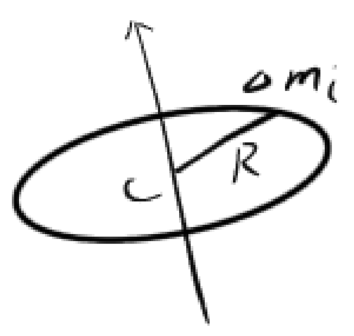
$\lambda = \frac{m}{L}$, $dm = \lambda dx$

$I_o = \int x^2 dm = \int_0^L x^2 dx \lambda = \frac{1}{3} mL^2$



$I_c = \int x^2 dm = \int_{-\frac{L}{2}}^{\frac{L}{2}} x^2 dx \lambda = \frac{1}{12} mL^2$

环



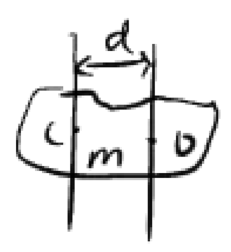
$I_c = \Sigma \Delta m_i R^2 = R^2 \Sigma \Delta m_i$

$I_c = mR^2$

盘: $dI = dm\, r^2$, $dm = \sigma\, 2\pi r\, dr$, $\sigma = \frac{m}{\pi R^2}$

$I = \int r^2 dm = \int_0^R r^2 \sigma\, 2\pi r\, dr = \frac{1}{2} mR^2$

平行轴定理:



$J_o = J_c + md^2$

证: 建2个坐标系, r为坐标表示

- 角动量: $F = \frac{dp}{dt} \to \tau = r \times \frac{dp}{dt} = \frac{d}{dt}(r \times p) - \underbrace{\frac{dr}{dt} \times p}_{0}$

$\to L = r \times p = Iw$, $\Sigma \tau = \frac{dL}{dt}$

$\Sigma \tau = 0$, $L = \text{constant}$

$F = ma = \frac{dp}{dt}$

$\tau = r \times \frac{dp}{dt} = I\alpha$

$L = r \times p = Iw$

- 自由度: $s = N - l$

广义坐标: $r = r(q_1, q_2, \ldots, q_s)$

手腕固定, 手具有19个自由度

- 拉格朗日方程

$L = K - U$, Example: for rotation in vertical plane:

$$L = \frac{1}{2} I \dot{\theta}^2 - mgh\sin\theta$$

- 哈密顿力学

广义动量: $p_i = \frac{\partial L}{\partial \dot{q}_i}$ 类$\left(\frac{dKv^2}{d\dot{x}}\right)$?

哈密顿正则方程:
$$\begin{cases} \dot{q}_i = \frac{\partial H}{\partial p_i} \\ \dot{p}_i = -\frac{\partial H}{\partial q_i} \end{cases}$$

$L = K - U$, $\sum_i p_i \dot{q}_i = 2K \longrightarrow H = K + U$

类($mv \cdot \dot{x} = mv^2 = 2K$)?

- $F = k\frac{q_1 q_2}{r^2} d\hat{r} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}\hat{r}$

$E = F/q$, $E = \int \frac{1}{4\pi\varepsilon_0}\frac{dq}{r^2}\hat{r}$, $dq = \begin{cases} \lambda L \\ \sigma A \\ \rho V \end{cases}$

$Q = \frac{dV}{dt} = A\bar{v} = \bar{v}\cdot A$

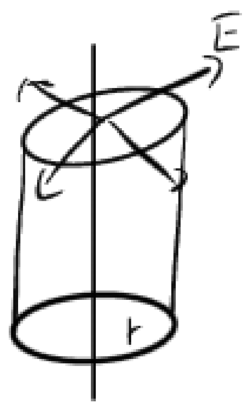
$d\Phi = E\cdot dA$, $\Phi_E = \oint_S \vec{E}\cdot d\vec{A} = \frac{1}{\varepsilon_0}\sum_{i=1}^{n} q_i^{(in)}$

均匀带电球面
任意点场强: (1) $0 < r < R$ $\oint_S \vec{E}\cdot d\vec{S} = 0$, $\vec{E} = 0$

(2) $r > R$

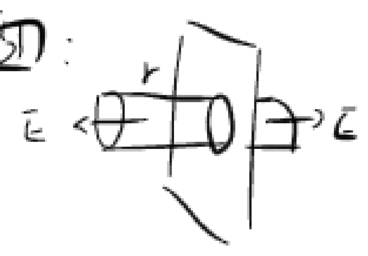
$$\oint_S \vec{E}\cdot d\vec{S} = \oint_S E\, dS = E\oint_S dS = E\cdot 4\pi r^2 = \frac{Q}{\varepsilon_0}$$

无限长均匀带电直线
附近电场强度:



$$\oint_S \vec{E}\cdot d\vec{S} = \overset{上}{0} + \overset{下}{0} + E\cdot 2\pi r h = \frac{\lambda h}{\varepsilon_0}$$

无限大均匀带电面:



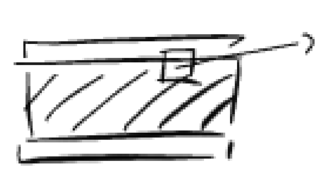
$$\oint_S \vec{E}\cdot d\vec{S} = 0 + E\cdot 2S = \frac{\sigma S}{\varepsilon_0}$$

- 导体内无电场, 无净电荷, 净电荷只分布于表面上

$\vec{E} = \frac{\sigma}{\varepsilon_0}\vec{e}_n$ 导体表面电荷面密度

- 电介质

$E_0 = \frac{\sigma}{\varepsilon_0}$, $U = \frac{U_0}{\varepsilon_r} \Rightarrow E = \frac{E_0}{\varepsilon_r}$



下表面所取高斯面

ε_r: 相对介电常数 , $\varepsilon_r \geq 1$

$\varepsilon = \varepsilon_0\varepsilon_r$ 电容率

充满 ε_r 的各向同性均匀电介质的平行板电容器

$E_0 = \frac{\sigma_0}{\varepsilon_0}$, $E' = \frac{\sigma'}{\varepsilon_0}$ (极化产生于电介质表面)

电介质内部电场强度

$\vec{E} = \vec{E}_0 + \vec{E}'$, $E = \frac{\sigma_0}{\varepsilon_0} - \frac{\sigma'}{\varepsilon_0} = \frac{\sigma_0}{\varepsilon_0\varepsilon_r}$ 因为 $E = \frac{E_0}{\varepsilon_r} = \frac{\sigma_0}{\varepsilon_0\varepsilon_r}$

$\Rightarrow \sigma' = \left(1 - \frac{1}{\varepsilon_r}\right)\sigma_0$, $Q' = \frac{\varepsilon_r - 1}{\varepsilon_r}Q_0$

- 有介质时的高斯

不包括极化电荷了

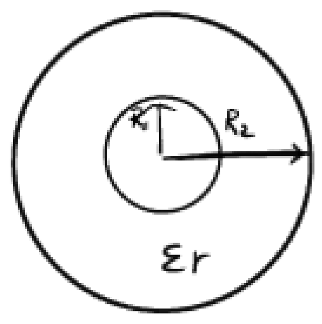
加入电介质(ε_r) $\oint_S \vec{E}\cdot d\vec{S} = \frac{1}{\varepsilon_0}\sum q_i = \frac{1}{\varepsilon_0}(\sigma_0 - \sigma')\Delta S = \frac{\sigma_0\Delta S}{\varepsilon_0\varepsilon_r} = \frac{1}{\varepsilon_0\varepsilon_r}\sum_i Q_{0i}$

$\oint_S \varepsilon_0\varepsilon_r\vec{E}\cdot d\vec{S} = \sum_i Q_{0i} \longrightarrow$ 自由电荷的代数和

电位移 $\vec{D} = \varepsilon_0\varepsilon_r\vec{E} = \varepsilon\vec{E}$, $\oint_S \vec{D}\cdot d\vec{S} = \sum_i Q_{0i}$

在真空中：$\varepsilon_r=1$，$\varepsilon=\varepsilon_0$，$\vec{D}=\varepsilon_0\vec{E}$，$\oint_S\vec{E}\cdot d\vec{S}=\frac{1}{\varepsilon_0}\sum_i q_{0i}$

例：$\oint_S\vec{D}\cdot d\vec{S}=\sum_i q_{0i}$，$\vec{D}=\varepsilon_0\varepsilon_r\vec{E}=\varepsilon\vec{E}$



$\oint\vec{D}\cdot d\vec{S}=\oint D\,dS=D\oint dS$　　$D\cdot4\pi r^2=\sum_i q_{0i}$

(1) $r<R_1$

$D\cdot4\pi r^2=0$，$D=0$，$E=0$

(2) $R_1<r<R_2$

$D\cdot4\pi r^2=\sum_i q_{0i}=Q$

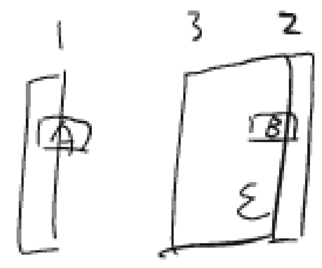
$D=\frac{Q}{4\pi r^2}$，$E=\frac{Q}{4\pi\varepsilon r^2}$

(3) $r>R_3$

$D\cdot4\pi r^2=\sum_i q_{0i}=Q$

$D=\frac{Q}{4\pi r^2}$，$E=\frac{Q}{4\pi\varepsilon_0 r^2}$

例：$\oint_S\vec{D}\cdot d\vec{S}=\sum_i q_{i0}$　$\oint_S\vec{D}\cdot d\vec{S}=D\Delta S$

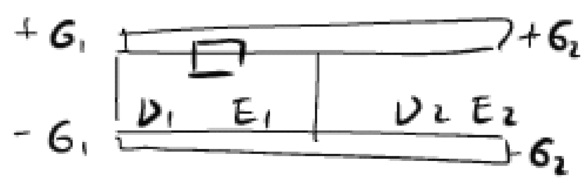


$D\Delta S=\sigma\Delta S$，$\Rightarrow D=\sigma$.

$E_A=\frac{\sigma}{\varepsilon_0}$，$E_B=\frac{\sigma}{\varepsilon}$（由上可知，只用管自由电荷）

$U_{12}=\int_1^2\vec{E}\cdot d\vec{l}=\int_1^3\vec{E}\cdot d\vec{l}+\int_3^2\vec{E}\cdot d\vec{l}=E_A(d-d_1)+E_Bd_1$

例：$U=Ed$，$U_0=E_0d=\frac{\sigma_0}{\varepsilon_0}d$



插入一半介质
保持电量不变

等势体 $U_1=E_1d$，$U_2=E_2d$ $\Rightarrow E_1=E_2$

$\oint_S\vec{D_1}\cdot d\vec{S}=D_1\Delta S=\sigma_1\Delta S$，$D_1=\sigma_1$，$E_1=\frac{\sigma_1}{\varepsilon_0\varepsilon_r}$

同理．$D_2=\sigma_2$，$E_2=\frac{\sigma_2}{\varepsilon_0}$

$E_1=E_2\Rightarrow\sigma_2=\frac{\sigma_1}{\varepsilon_r}$

电荷守恒　$\sigma_1\cdot\frac{S}{2}+\sigma_2\cdot\frac{S}{2}=\sigma_0S$

$\sigma_1=\frac{5}{3}\sigma_0$，$\sigma_2=\frac{1}{3}\sigma_0$

$E=E_1=E_2=\frac{\sigma_2}{\varepsilon_0}=\frac{2\sigma_0}{\varepsilon_0(1+\varepsilon_r)}=\frac{1}{3}E_0$

$Q/V=C$，$E=\frac{V}{d}A=\frac{Q}{\varepsilon}$

$E=\begin{cases}0 & (r<R)\\ \frac{Q}{4\pi\varepsilon_0r^2} & (r>R)\end{cases}$

$V=\int_R^\infty\vec{E}\cdot d\vec{r}=\int_R^\infty\frac{Q}{4\pi\varepsilon_0r^2}dr=\frac{Q}{4\pi\varepsilon_0R}$

$C=\frac{Q}{V}=4\pi\varepsilon_0R$

$C=\frac{\varepsilon A}{d}=\frac{\varepsilon_rA}{4\pi kd}$，$k=\frac{1}{4\pi\varepsilon_0}$

$I=\frac{dq}{dt}=\frac{nsdlq}{dl/v}=qnvs$

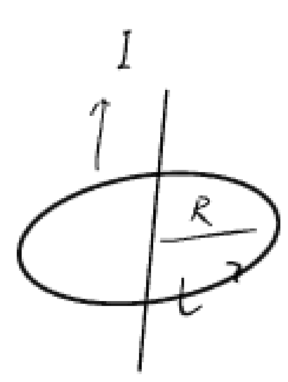
$I=\int j\,dA$（电流密度）

$\vec{E}_k=\frac{\vec{F}_k}{q}$　非静电力场强

$\varepsilon=\frac{W}{q}=\oint\vec{E}_k\cdot dl=\int_{-(in)}^{+}\vec{E}_k\cdot d\vec{l}$　电动势

$dF=IdI\times B$

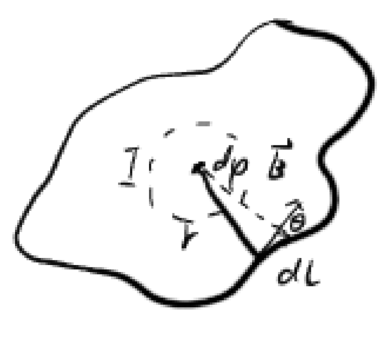
$B=\frac{\mu_0I}{2\pi r}$



同心圆　$\oint_L\vec{B}\cdot d\vec{l}=\oint_L B\,dl=\oint\frac{\mu_0I}{2\pi R}dl=\mu_0I$

任意形状环路　r处，$B=\frac{\mu_0I}{2\pi r}$

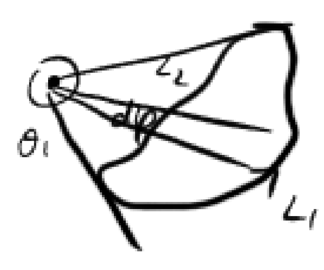
$\oint_L\vec{B}\cdot d\vec{l}=\oint B\,dl\cos\theta=\oint Br\,d\varphi=\int_0^{2\pi}\frac{\mu_0I}{2\pi}d\varphi=\mu_0I$



环路不包围直导线

$\oint_L\vec{B}\cdot d\vec{l}=\int_{L_1}\vec{B}\cdot d\vec{l}+\int_{L_2}\vec{B}\cdot d\vec{l}$

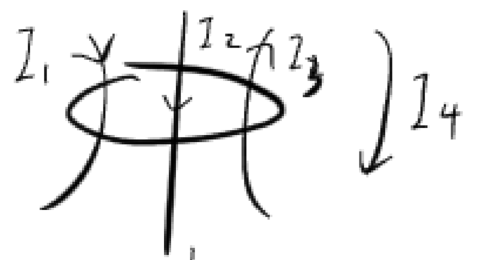
$=\int_0^{\theta_1}\frac{\mu_0I}{2\pi}d\varphi+\int_{\theta_1}^{0}\frac{\mu_0I}{2\pi}d\varphi=0$



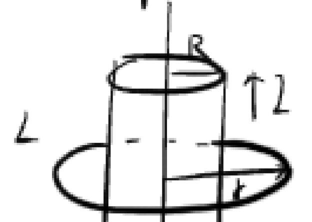
· $\oint_L\vec{B}\cdot d\vec{l}=\oint_L\sum_i\vec{B_i}\cdot d\vec{l}=\sum_i\oint_L\vec{B_i}\cdot d\vec{l}=\sum_i\mu_0I_{\underline{\text{内}i}}$　回路所包含的电流

安培环路定理　$\oint_L\vec{B}\cdot d\vec{l}=\mu_0\sum_{i=1}I_{\text{内}i}$

例：I_1 I_2 I_3 I_4



$\oint_L \vec{B}\cdot d\vec{l} = \mu_0(-I_1 - I_2 + I_3)$



拉直

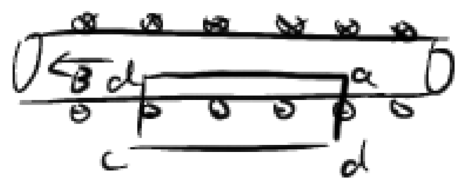
$\oint_L \vec{B}\cdot d\vec{l} = \oint_L B\,dl = B2\pi r$

$r<R$　　$B2\pi r = 0$，$B=0$

$r>R$　　$B2\pi r = \mu_0 I$　　$B = \frac{\mu_0 I}{2\pi r}$

（圆柱　$r<R$　　$B2\pi r = \mu_0 \frac{I}{\pi R^2}\pi r^2$ ）

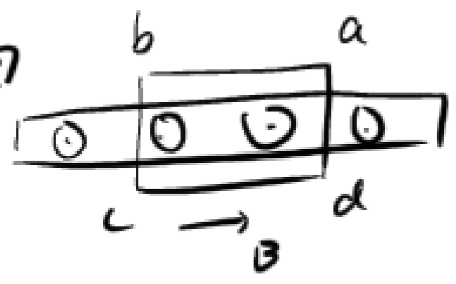
螺线管



$\oint_L \vec{B}\cdot d\vec{l} = \int_a^b \vec{B}\cdot d\vec{l} + \underbrace{\int_b^c \vec{B}\cdot d\vec{l} + \int_d^a \vec{B}\cdot d\vec{l}}_{点乘垂直为0} + \underbrace{\int_c^d \vec{B}\cdot d\vec{l}}_{外部B\to 0} = B\overline{ab}$

$B\overline{ab} = \mu_0 n \overline{ab} I$　　　$B = \mu_0 n I$

均匀载流平面



$\oint_L \vec{B}\cdot d\vec{l} = \int_a^b \vec{B}\cdot d\vec{l} + \int_b^c \vec{B}\cdot d\vec{l} + \int_c^d \vec{B}\cdot d\vec{l} + \int_d^a \vec{B}\cdot d\vec{l} = 2B\overline{ab}$

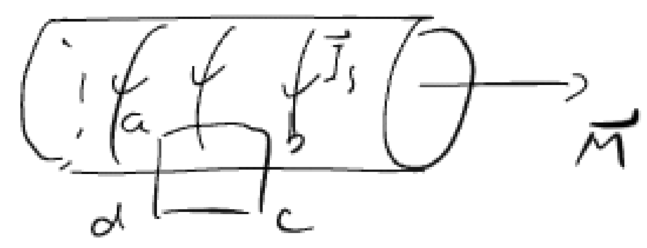
$B\cdot 2\overline{ab} = \mu_0 i \overline{ab}$　　　$B = \frac{\mu_0 i}{2}$　（i 为线密度）

- 磁化面电流密度　$j_s = \frac{I_s}{L}$

磁化强度　$\vec{M} = \frac{\sum \vec{m}_分}{\Delta V}$

$|\vec{M}| = \left|\frac{\sum \vec{m}_分}{\Delta V}\right| = \frac{I_s S}{L S} = j_s$　　数值上等于磁化面电流密度

$\oint_L \vec{M}\cdot d\vec{l} = \int_a^b \vec{M}\cdot d\vec{l} + \int_b^c \vec{M}\cdot d\vec{l} + \int_c^a \vec{M}\cdot d\vec{l} + \int_d^a \vec{M}\cdot d\vec{l}$



$= \int_a^b \vec{M}\cdot d\vec{l} = \int_a^b M\,dl = M\overline{ab} = I_s(ab)$

磁化强度沿任意闭合回路的环流，等于穿过回路所包围的磁化电流

磁介质中的安培环路定理

$\oint_L \vec{B}\cdot d\vec{l} = \mu_0 \sum I_{内i}$　　真空中

$\oint_L \vec{B}\cdot d\vec{l} = \mu_0 \sum (I_c + I_s) = \mu_0 \sum I_c + \mu_0 \oint_L \vec{M}\cdot d\vec{l}$　　介质

$\oint_L (\frac{\vec{B}}{\mu_0} - \vec{M})\cdot d\vec{l} = \sum I_c$

$\to \oint_L (\frac{\vec{B}}{\mu_0} - \vec{M})\cdot d\vec{l} = \sum I_c$

令 $\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$ —— 磁场强度

$\oint_L \vec{H}\cdot d\vec{l} = \sum I_c$ —— 介质中的环路定理（较广情况）

$\vec{M} = k\vec{H}$　　　k —— 磁化率

$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M} = \frac{\vec{B}}{\mu_0} - k\vec{H} \Rightarrow \vec{B} = \mu_0(1+k)\vec{H}$

$\vec{B} = \mu_0\mu_r\vec{H} = \mu\vec{H}$　　$\begin{cases}\mu_r = 1+k & 相对磁导率\\ \mu = \mu_0\mu_r & 磁导率\end{cases}$

例：环状螺线管

$\oint_L \vec{H}\cdot d\vec{l} = H\cdot 2\pi r = n 2\pi r I_0$

$H = nI_0$，　$B = \mu H = \mu n I_0$

全电流安培定律

$I_D = I \to \frac{dQ}{dt} = \frac{d(CV)}{dt}$

$= \frac{d}{dt}(\frac{\varepsilon A}{d} E d)$

$I_D = \varepsilon \frac{d(EA)}{dt} = \frac{d}{dt}(DA)$

$\oint H\,dl = I_{in} + \int \frac{\partial D}{\partial t}\cdot dA$

$= \int (j + \frac{\partial D}{\partial t})\cdot dA$

旋度

$\int_a^b H\,dl + \int_c^d H\,dl = H_x(y_0)dx - H_x(y_0+dy)dx$

$= -\frac{\partial H_x}{\partial y}dy\,dx = -\frac{\partial H_x}{\partial y}dA$

$\oint H\cdot dl = (-\frac{\partial H_x}{\partial y} + \frac{\partial H_y}{\partial x})dA = (j_z + \frac{\partial D_z}{\partial t})dA$

$(-\frac{\partial H_x}{\partial y} + \frac{\partial H_y}{\partial x})k = (j_z + \frac{\partial D_z}{\partial t})k$

$\to \nabla \times H = j + \frac{\partial D}{\partial t}$

$\nabla \times J = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ J_x & J_y & J_z \end{vmatrix}$

- 法拉第

$\varepsilon_i = -k\frac{d\Phi_m}{dt}$　　　$k=1$．ε_i 伏特，Φ_m 韦伯

$\varepsilon_i = -\frac{d\Phi_m}{dt}$

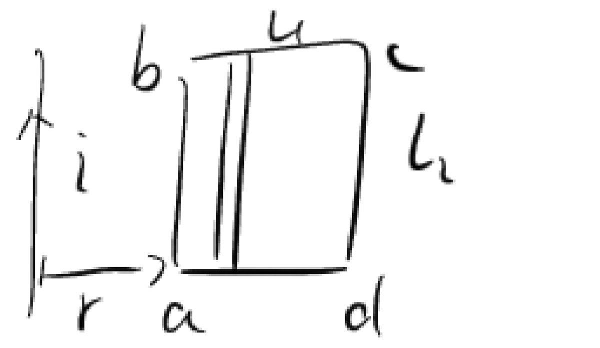
$\Psi = \sum_{i=1}^{N} \Phi_{mi}$　　磁通链数（N匝串连）　　$\Psi = N\Phi_m$　　N匝完全相同线圈串联

$\varepsilon_i = -\frac{d\Psi}{dt}$　　　　$\varepsilon_i = -N\frac{d\Phi_m}{dt}$

感应电动势计算

例：

$i = I_0 \sin\omega t$

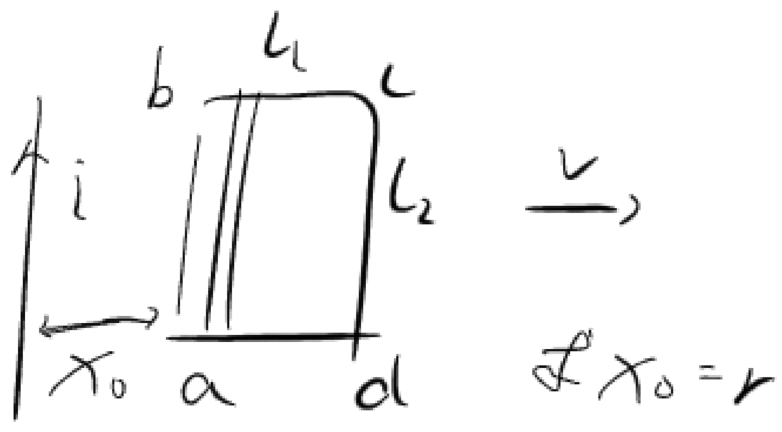


$B = \frac{\mu_0 i}{2\pi x}$, $dS = l_2 dx$

$\Phi_m = \int_s \vec{B}\cdot d\vec{S} = \int_r^{r+l_1} \frac{\mu_0 i}{2\pi x} l_2 dx = \frac{\mu_0 I_0 l_2}{2\pi}(\sin\omega t)\ln\frac{r+l_1}{r}$

$\varepsilon_i = -\frac{d\Phi_m}{dt} = -\frac{\mu_0 I_0}{2\pi} l_2 \omega(\cos\omega t)\ln\frac{r+l_1}{r}$

例.



$B = \frac{\mu_0 I}{2\pi x}$, $dS = l_2 dx$

$dx/dt = v$

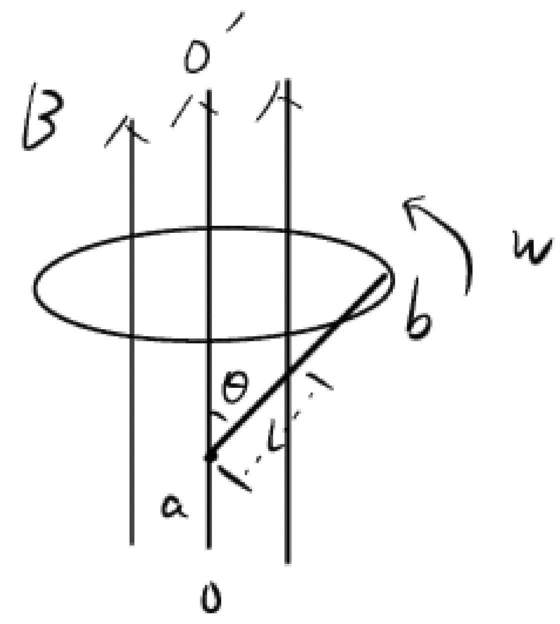
$x_0 = r$

动生电动势

$\vec{F}_k = q\vec{v}\times\vec{B}$, $\vec{E}_k = \frac{\vec{F}_k}{q}$, $\vec{E}_k = \vec{v}\times\vec{B}$ $\quad$ $\varepsilon_i = \int \vec{E}\cdot d\vec{l}$ (基本定义)

$\varepsilon_i = \int_l (\vec{v}\times\vec{B})\cdot d\vec{l}$

例 取线元矢量 $d\vec{l}$, $v = r\omega = l\omega\sin\theta$



$\vec{E}_k = \vec{v}\times\vec{B}$

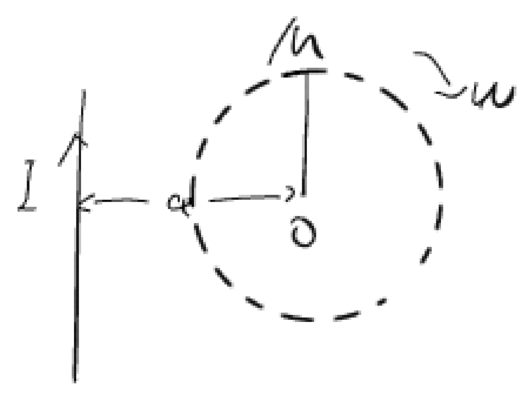
$E_k = vB = l\omega B\sin\theta$

$d\varepsilon_i = \vec{E}_k\cdot d\vec{l}$

$d\varepsilon_i = E_k\, dl\cos(90^\circ-\theta) = E_k\, dl\sin\theta = l\omega B\, dl\sin^2\theta$

$\varepsilon_i = \int_L \vec{E}_k\cdot d\vec{l} = \int_0^L l\omega B\, dl\sin^2\theta = \frac{1}{2}\omega B L^2\sin^2\theta$

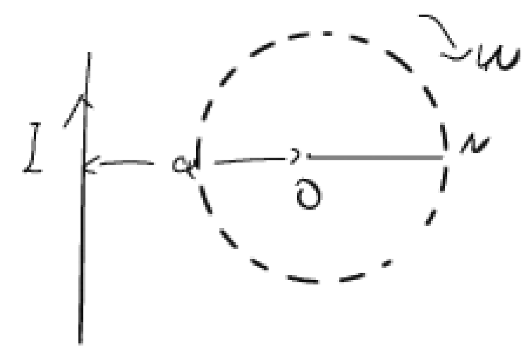
例： 1) $B = \frac{\mu_0 I}{2\pi a}$, $v = l\omega$, $\vec{E}_k = \vec{v}\times\vec{B}$,



$E_k = vB = l\omega\frac{\mu_0 I}{2\pi a}$, $d\varepsilon_i = E_k\, dl = \frac{\mu_0 I\omega}{2\pi a} l\, dl$

$\varepsilon_i = \int_L \vec{E}_k\cdot d\vec{l} = \int_0^L \frac{\mu_0 I\omega}{2\pi a} l\, dl = \frac{\mu_0 I\omega}{4\pi a}L^2$

2) $B = \frac{\mu_0 I}{2\pi(a+x)}$, $v = x\omega$, $\vec{E}_k = \vec{v}\times\vec{B}$



$E = vB = \frac{\mu_0 I\omega}{2\pi}\frac{x}{(x+a)}$, $d\varepsilon_i = E_k\, dx = \frac{\mu_0\omega x}{2\pi(x+a)}dx$

$\varepsilon_i = \int_0^L \frac{\mu_0 I\omega}{2\pi}\frac{x}{(x+a)}dx = \frac{\mu_0 I\omega}{2\pi}\left[L - a\ln\left(\frac{a+L}{a}\right)\right]$

感生电动势和感生电场

电动势定义 $\varepsilon_i = \oint_L \vec{E}_{感}\cdot d\vec{l}$, 法拉第 $\varepsilon_i = -\frac{d\Phi_m}{dt}$

$\Rightarrow \oint_L \vec{E}_{感}\cdot d\vec{l} = -\frac{d\Phi_m}{dt}$

$\frac{d\Phi_m}{dt} = \frac{d}{dt}\int_s \vec{B}\cdot d\vec{S} = \int_s \frac{d\vec{B}}{dt}\cdot d\vec{S}$

$\varepsilon_i = \oint_L \vec{E}_{感}\cdot d\vec{l} = -\int_s \frac{d\vec{B}}{dt}\cdot d\vec{S}$

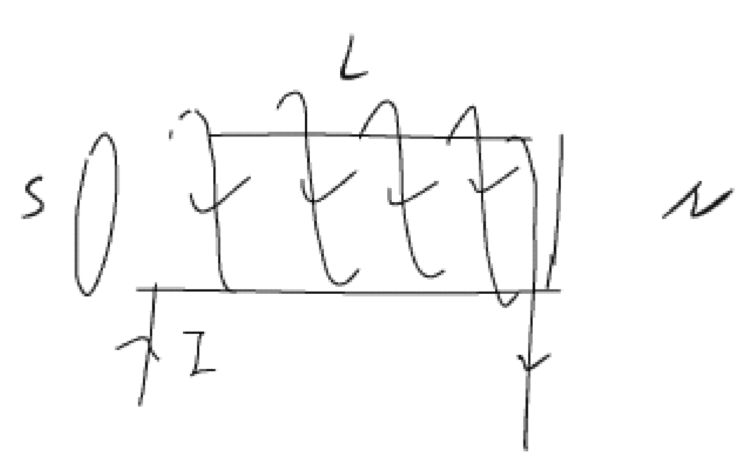
自感.

系数 $\Psi = LI$ $\quad$ $L = \frac{\Psi}{I}$

$\varepsilon_L = -\frac{d\Psi}{dt} = -\frac{d}{dt}(LI) = -L\frac{dI}{dt} - I\underbrace{\frac{dL}{dt}}_{=0}$ 定量

$\Rightarrow \varepsilon_L = -L\frac{dI}{dt}$

例：



$B = \mu n I = \mu\frac{N}{l}I$

$\Psi = NBS = \mu\frac{N^2}{l}SI$

$L = \frac{\Psi}{I} = \mu\frac{N^2}{l}S = \mu n^2 V$

- 互感

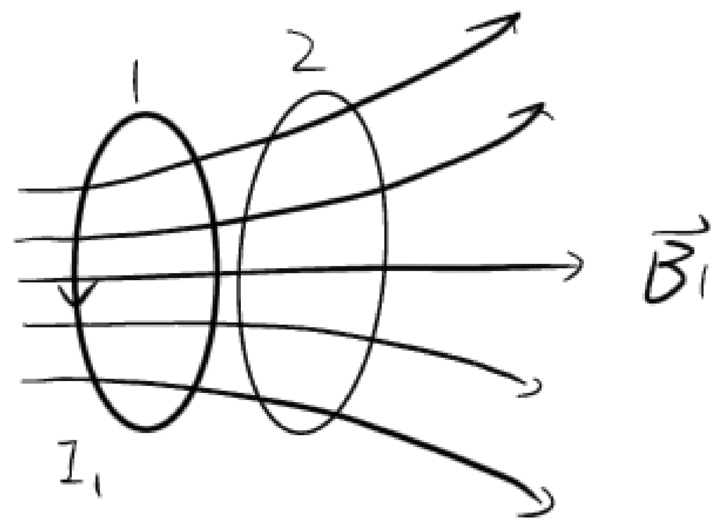
系数 $I_1(t)\to B_1(t)\to \Psi_{21}(t)\to \varepsilon_{21}$,

$\Psi_{21}\propto B_1\propto I_1$, $\Psi_{21}=M_{21}I_1$

$\varepsilon_{21}=-\frac{d\Psi_{21}}{dt}=-M_{21}\frac{dI_1}{dt}$

相反同理 $\varepsilon_{12}=-M_{12}\frac{dI_2}{dt}$

$\Rightarrow M_{12}=M_{21}=M\longrightarrow$ 互感系数 $M=\frac{\Psi_{21}}{I_1}$



- 磁矢势 A_m 磁场的散度为零, $\nabla\cdot B=0$ （散度: $\nabla\cdot F=\frac{\partial F_x}{\partial x}+\frac{\partial F_y}{\partial y}+\frac{\partial F_z}{\partial z}$）

$\nabla\cdot E=\frac{\rho}{\varepsilon}\to\nabla\cdot D=\rho$

定义 $B=\nabla\times A_m$

高斯 $\Phi_m=\int_S B\cdot dA=\oint_{\partial S}A_m\cdot dl=\int_S(\nabla\times A_m)\cdot dA$

$dA_m=\frac{\mu_0 I\,dl}{4\pi r}$, $dV=\frac{1}{4\pi\varepsilon}\cdot\frac{dq}{r}$ 电势

$\oint E\cdot dl=-\frac{d}{dt}\int B\cdot dA\Rightarrow\nabla\times E=-\frac{\partial B}{\partial t}=-\frac{\partial}{\partial t}(\nabla\times A_m)=\nabla\times\left(-\frac{\partial A_m}{\partial t}\right)$

$E=-\frac{\partial A_m}{\partial t}+\nabla\phi$

例: $B=\nabla\times AM=\nabla\times\ln x\, j$

$$\begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & \ln x & 0 \end{vmatrix}$$

$i\times 0+j\times 0+k\times\frac{\partial(\ln x)}{\partial x}$

- 洛伦兹力 （合力）

$F=qE+qv\times B=q(E+v\times B)$

- $u=\frac12\vec{D}\cdot\vec{E}$ $U=\frac12CV^2$

$u=\frac12\vec{B}\cdot\vec{H}$ $U=\frac12LI^2$

- $S=u\vec{v}=\vec{E}\times\vec{H}$

- $\nabla\cdot\vec{E}=\frac{\rho}{\varepsilon_0}$ $\nabla\cdot\vec{B}=0$

$\nabla\times\vec{E}=-\frac{\partial\vec{B}}{\partial t}$ $\nabla\times\vec{B}=\varepsilon_0\mu_0\frac{\partial\vec{E}}{\partial t}+\mu_0 J_e$

真空 $\Rightarrow$ $\nabla\cdot\vec{E}=0$ $\nabla\cdot\vec{B}=0$

$\nabla\times\vec{E}=-\frac{\partial\vec{B}}{\partial t}$ $\nabla\times\vec{B}=\frac{1}{c^2}\frac{\partial\vec{E}}{\partial t}$

- 无散场, 无旋场

$\nabla\cdot E=0$, $\nabla\times E=0$

$\nabla\times(\nabla\times\vec{A})=\nabla(\nabla\cdot\vec{A})-\nabla^2\vec{A}$

$\iint_S(\nabla\times F)\cdot d\vec{s}=\oint_L F\cdot d\vec{l}$

$\iiint_V(\nabla\cdot F)\,dV=\oiint_S F\cdot d\vec{s}$

$\nabla\left(\frac1r\right)=-\frac{\vec{r}}{r^3}$

$\nabla\cdot\vec{B}=0$

$\vec{B}=\nabla\times\vec{A}$

$\vec{E}=-\nabla\varphi-\frac{d\vec{A}}{dt}$

求散, 并 $\nabla\cdot\vec{E}=\frac{\rho}{\varepsilon_0}$

$\nabla^2\varphi+\frac{\partial}{\partial t}\nabla\cdot\vec{A}=-\frac{\rho}{\varepsilon_0}$

$\nabla\times\vec{B}=\frac{1}{c^2}\frac{\partial\vec{E}}{\partial t}+\mu_0 J_e$ $\leftarrow$ $\frac{\partial\vec{E}}{\partial t}=-\frac{1}{c^2}\frac{\partial}{\partial t}(\nabla\varphi)-\frac{1}{c^2}\frac{\partial^2\vec{A}}{\partial t^2}$

$\nabla\times(\nabla\times\vec{A})=\nabla(\nabla\cdot\vec{A})-\nabla^2\vec{A}=\frac{1}{c^2}\frac{\partial\vec{E}}{\partial t}+\mu_0 J_e$

$=\mu_0 J-\frac{1}{c^2}\frac{\partial}{\partial t}(\nabla\varphi)-\frac{1}{c^2}\frac{\partial^2\vec{A}}{\partial t^2}$

$\nabla^2\vec{A}-\frac{1}{c^2}\frac{\partial^2\vec{A}}{\partial t^2}-\nabla\left(\nabla\cdot\vec{A}+\frac{1}{c^2}\frac{\partial\varphi}{\partial t}\right)=-\mu_0 J$

$\nabla\cdot\vec{A}=0$ 库伦规范

$\downarrow$

$$\nabla^2\vec{A}-\frac{1}{c^2}\frac{\partial^2\vec{A}}{\partial t^2}-\frac{1}{c^2}\frac{\partial\varphi}{\partial t}=-\mu_0\vec{J}$$

$$\nabla^2\varphi=-\frac{\rho}{\varepsilon_0}$$

$\nabla\cdot\vec{A}+\frac{1}{c^2}\frac{\partial\varphi}{\partial t}=0$ 劳规范

$\downarrow$

$$\nabla^2\vec{A}-\frac{1}{c^2}\frac{\partial^2\vec{A}}{\partial t^2}=-\mu_0\vec{J}$$

$$\nabla^2\varphi-\frac{1}{c^2}\frac{\partial^2\varphi}{\partial t^2}=-\frac{\rho}{\varepsilon_0}$$

- $\nabla^2\vec{E}-\frac{1}{c^2}\frac{\partial^2\vec{B}}{\partial t^2}=0$ $\quad$ $\nabla^2\vec{B}-\frac{1}{c^2}\frac{\partial^2\vec{E}}{\partial t^2}=0$

$e^{i\theta}=\cos\theta+i\sin\theta$ $\quad$ 真空中麦~方程的意义(电磁波)

- 时谐电磁波

$\vec{E}(\vec{r},t)=\vec{E}(\vec{r})e^{-i\omega t}$ $\quad$ $\vec{B}(\vec{r},t)=\vec{B}(\vec{r})e^{-i\omega t}$

$\nabla\times\vec{E}=-\frac{\partial\vec{B}}{\partial t}\longrightarrow\nabla\times\vec{E}=i\omega\vec{B}$

$\nabla\times\vec{B}=\frac{1}{c^2}\frac{\partial\vec{E}}{\partial t}\longrightarrow\nabla\times\vec{B}=-i\omega\vec{E}$

$\nabla\cdot\vec{E}=0\longrightarrow\nabla\cdot\vec{E}=0$

$\nabla\cdot\vec{B}=0\longrightarrow\nabla\cdot\vec{B}=0$

- $\nabla^2\vec{E}+k^2\vec{E}=0$, $\quad\nabla^2\vec{B}+k^2\vec{B}=0$, $\quad k=\frac{\omega}{c}$

$\vec{B}=-\frac{i}{\omega}\nabla\times\vec{E}$, $\quad\vec{E}=\frac{i}{kc}\nabla\times\vec{B}$

- 平面电磁波是一种特殊的时谐。

$$\vec{E}(\vec{r},t)=\vec{E}_0e^{i(\vec{k}\cdot\vec{r}-\omega t)}=\vec{E}_0e^{i\vec{k}\vec{r}}e^{-i\omega t}$$

$$\vec{B}(\vec{r},t)=\vec{B}_0e^{i(\vec{k}\cdot\vec{r}-\omega t)}=\vec{B}_0e^{i\vec{k}\vec{r}}e^{-i\omega t}$$

特点: $\frac{\partial}{\partial t}\longrightarrow-i\omega$

$\vec{B}=\vec{B}_0e^{i(\vec{k}\cdot\vec{r}-\omega t)}\longrightarrow\nabla\cdot\vec{B}=0\longrightarrow\vec{k}\cdot\vec{B}=0$

$\vec{E}=\vec{E}_0e^{i(\vec{k}\cdot\vec{r}-\omega t)}\longrightarrow\nabla\cdot\vec{E}=0\longrightarrow\vec{k}\cdot\vec{E}=0$

$\vec{B}_0 e^{i(\vec{k}\cdot\vec{r}-\omega t)}$ $\vec{E}_0 e^{i(\vec{k}\cdot\vec{r}-\omega t)}$

$$\nabla\times\vec{E} = -\frac{d\vec{B}}{dt}$$
$$\downarrow$$
$$\vec{k}\times\vec{E} = \omega\vec{B}$$
$$\downarrow$$
$$\vec{B} = \frac{1}{\omega}\vec{k}\times\vec{E}$$
$$\downarrow$$
电场与磁场垂直

$\vec{B}_0 e^{i(\vec{k}\cdot\vec{r}-\omega t)}$ $\vec{E}_0 e^{i(\vec{k}\cdot\vec{r}-\omega t)}$

$$\nabla\times\vec{B} = \frac{1}{c^2}\frac{\partial\vec{E}}{\partial t}$$
$$\downarrow$$
$$\vec{k}\times\vec{B} = -\frac{\omega}{c^2}\vec{E}$$
$$\downarrow$$
$$k^2\vec{B} = \frac{\omega^2\vec{B}}{c^2}$$
$$\downarrow$$
$$k = \frac{\omega}{c} = \omega\sqrt{\mu\varepsilon} \qquad c^2 = \frac{1}{\mu\varepsilon_0}$$

真空中，k与ω相关系

- $\vec{A}$和φ与$\vec{B}$和$\vec{E}$

洛伦兹规范：$\nabla^2\vec{A} - \frac{1}{c^2}\frac{\partial^2\vec{A}}{\partial t^2} = 0$ $\quad \nabla^2\varphi - \frac{1}{c^2}\frac{\partial^2\varphi}{\partial t^2} = 0$

$$\vec{A} = \vec{A}_0 e^{i(\vec{k}\cdot\vec{r}-\omega t)} \qquad \varphi = \varphi_0 e^{i(\vec{k}\cdot\vec{r}-\omega t)}$$

$$\nabla\cdot\vec{A} + \frac{1}{c^2}\frac{\partial\varphi}{\partial t} = 0$$
$$\downarrow$$
$$\varphi_0 = \frac{c^2}{\omega}\vec{k}\cdot\vec{A}_0 \rightarrow \varphi = \frac{c^2}{\omega}\vec{k}\cdot\vec{A}$$
$$\downarrow$$
$$\vec{E} = -\nabla\varphi - \frac{\partial\vec{A}}{\partial t} = -i\vec{k}\varphi + i\omega\vec{A} \rightarrow \vec{B} = -\frac{i}{\omega}\vec{k}\times\vec{A}$$

横波：$\vec{k}\cdot\vec{A} = 0 \rightarrow \varphi_0 = \frac{c^2}{\omega}\vec{k}\cdot\vec{A}_0 \rightarrow \varphi = 0 \rightarrow \vec{E} = -\nabla\varphi - \frac{\partial\vec{A}}{\partial t} = -i\vec{k}\varphi + i\omega\vec{A}$

$$\downarrow$$
$$\vec{E} = i\omega\vec{A}, \quad \vec{B} = \frac{1}{\omega}\vec{k}\times\vec{E} = i\vec{k}\times\vec{A}$$

- $\vec{D}(\omega) = \varepsilon(\omega)\vec{E}(\omega)$ $\quad \vec{H}(\omega) = \mu(\omega)\vec{B}(\omega)$

$$v^2 = \frac{1}{\varepsilon(\omega)\mu(\omega)} = v^2(\omega) \qquad k = \frac{\omega}{v}\sqrt{1 + i\frac{\sigma}{\omega\varepsilon}}$$

$$\nabla^2\vec{E} + k^2\vec{E} = 0 \quad \underline{\nabla\cdot\vec{E} = 0} \quad \vec{B} = -\frac{i}{\omega}\nabla\times\vec{E}$$
$$\nabla^2\vec{B} + k^2\vec{B} = 0 \quad \underline{\nabla\cdot\vec{B} = 0} \quad \vec{E} = \frac{i}{k}\sqrt{\mu\varepsilon}\,\nabla\times\vec{B}$$

介质中的时谐形式上与真空中一样

$$k = \frac{\omega}{c} \qquad k = \frac{\omega}{v}\sqrt{1 + i\frac{\sigma}{\omega\varepsilon}}$$

- 谐振腔（一个金属腔）

$$\nabla^2 \vec{E} + k^2 \vec{E} = 0$$

$$\downarrow$$

$$\nabla_x^2 E_x + \nabla_y^2 E_y + \nabla_z^2 E_z + k_x^2 E_x + k_y^2 E_y + k_z^2 E_z = 0$$

$$\downarrow$$

$$\nabla_x^2 E_x + k^2 E_x = 0$$

$$\nabla_x^2 E_y + k^2 E_y = 0$$

$$\nabla_x^2 E_z + k^2 E_z = 0$$

$$\nabla_x^2 E_x(x,y,z) + k_x^2 E_x(x,y,z) = 0$$

E_x, E_y, E_z 统一写成 u

$$\nabla^2 u + k^2 u = 0$$

$$u(x,y,z) = X(x)Y(y)Z(z)$$

$$\frac{d^2 X}{dx^2} + k_x^2 X = 0 \quad \cdots$$

$$k_x^2 + k_y^2 + k_z^2 = w^2 \mu \varepsilon$$

$$u(x,y,z) = (C_1 \cos k_x x + D_1 \sin k_x x)(\cdots$$

边界条件，$\nabla \cdot \vec{E} = 0 = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}$

$$\downarrow$$

$$\frac{\partial E_n}{\partial n}\Big|_{n=0} = 0 = \frac{\partial E_n}{\partial n}\Big|_{n=L}$$

$$E_y|_{y=0} = E_z|_{z=0} = 0, \quad \frac{\partial E_x}{\partial x}\Big|_{x=0} =$$

$$\vdots$$

$$\downarrow$$

$$E_x = A_1 \cos k_x x \cdot \sin k_y y \cdot \sin k_z \cdot z$$

$$\vdots$$

再考虑，$\frac{\partial E_n}{\partial n}\Big|_{n=L} = 0$

$$\downarrow$$

$$k_x = \frac{m\pi}{L_1}, \quad k_y = \frac{n\pi}{L_2}, \quad k_z = \frac{p\pi}{L_3}$$

$$\downarrow$$

$$\omega_{mnp} = \frac{\pi}{\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m}{L_1}\right)^2+\left(\frac{n}{L_2}\right)^2+\left(\frac{p}{L_p}\right)^2}$$

$$\downarrow$$

$$\frac{m^2}{\left(\frac{\omega L_1}{\pi c}\right)^2} + \frac{n^2}{\left(\frac{\omega L_2}{\pi c}\right)^2} + \frac{p^2}{\left(\frac{\omega L_3}{\pi c}\right)^2} = 1$$

$$\downarrow$$

$$N = \frac{4}{3}\pi\left(\frac{\omega L_1}{c}\right)\cdots = \frac{4}{3}\pi\left(\frac{\omega}{c}\right)^3 L_1L_2L_3$$

$$\downarrow$$

$$n = \frac{4\pi}{3}\left(\frac{\omega}{c}\right)^3 \rightarrow dn = \frac{4\pi}{c^3}\omega^2 d\omega$$

- Snell's

$$\vec{E}_0 \cdot e^{i\vec{k}\cdot\vec{r}} \cdot e^{-i\omega t} + \vec{E}_0' e^{i\vec{k}'\cdot\vec{r}} \cdot e^{-i\omega' t} = \vec{E}_0'' e^{i\vec{k}''\cdot\vec{r}} \cdot e^{-i\omega'' t}$$

全时成立，$\omega = \omega' = \omega''$

相位相同要求 $\vec{k}\cdot\vec{r} = \vec{k}'\cdot\vec{r} = \vec{k}''\cdot\vec{r}$

$$\rightarrow \vec{E}_0 + \vec{E}_0' = \vec{E}_0''$$

$$\langle\psi|\psi\rangle = \int_{-\infty}^{+\infty} |\psi|^2 dx = 1$$

$$\psi = \psi(x, y, z, t)$$ 只有$|\psi|^2$与宏观关联

$$\psi = C_1\psi_1 + C_2\psi_2 + \cdots + C_n\psi_n = \sum_n C_n\psi_n$$

$|C_n|^2$ 为塌缩为本征态的 ψ_n 概率

$$|C_1|^2 + |C_2|^2 + \text{—} + |C_n|^2 = 1$$

$$|\psi\rangle = C_1|\underset{E_1}{\psi_1}\rangle + C_2|\underset{E_2}{\psi_2}\rangle + C_3|\psi_3\rangle + \cdots + C_n|\psi_n\rangle$$

$$\hat{A}|\psi_n\rangle = A_n|\psi_n\rangle$$

$$E \to \hat{A} \to i\hbar\frac{\partial}{\partial t}$$

$I + V$ $\qquad \frac{p^2}{2m} = \frac{(-i\hbar\frac{\partial}{\partial t})^2}{2m}$

$$i\hbar\frac{\partial}{\partial t}|\psi\rangle = \hat{A}|\psi\rangle$$

$\langle E\rangle = |C_1|^2 E_1 + \cdots + |C_n|^2 E_n$ 平均能量

$$|\psi\rangle = \begin{pmatrix} C_1 \\ C_2 \\ \vdots \\ C_n \end{pmatrix} \qquad (|\psi\rangle)^* = \begin{pmatrix} C_1^* \\ C_2^* \\ \vdots \\ C_n^* \end{pmatrix}$$

$$|\psi\rangle = |x\rangle\langle x|\psi\rangle = |\psi(x)\rangle, \quad |\phi\rangle = |\phi(x)\rangle$$

$$\langle\phi|\psi\rangle = \int \Phi^*(x)\psi(x)\,dx$$

$$|\psi\rangle\langle\phi|$$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \qquad |\phi\rangle = \gamma|0\rangle + \delta|1\rangle$$

$$\begin{aligned}\langle\phi|\psi\rangle &= (\alpha|0\rangle + \beta|1\rangle)^*(\gamma|0\rangle + \delta|1\rangle) \\ &= (\alpha^*\langle 0| + \beta^*\langle 1|)(\gamma|0\rangle + \delta|1\rangle) \\ &= \alpha^*\gamma + \beta^*\delta\end{aligned}$$

$$\begin{aligned}|\psi\rangle\langle\phi| &= (\alpha|0\rangle + \beta|1\rangle)(\gamma|0\rangle + \delta|1\rangle)^* \\ &= \begin{pmatrix} \alpha\gamma^*|0\rangle\langle 0| & \alpha\delta^*|0\rangle\langle 1| \\ \beta\gamma^*|1\rangle\langle 0| & \beta\delta^*|1\rangle\langle 1| \end{pmatrix}\end{aligned}$$

$$\hat{A}|\psi_n\rangle = A_n|\psi_n\rangle$$

$$\langle\psi_n|\hat{A}|\psi_n\rangle = A_n\langle\psi_n|\psi_n\rangle = A_n$$

$$\begin{aligned}\langle A\rangle = \langle\psi|\hat{A}|\psi\rangle &= \sum_m C_m^*\langle\psi_m|\hat{A}\sum_n C_n|\psi_n\rangle \\ &= \sum_m\sum_n C_m^* C_n\langle\psi_m|\psi_n\rangle A_n \\ &= \sum_n C_n^* C_n\langle\psi_n|\psi_n\rangle A_n \\ &= \sum_n |C_n|^2 A_n\end{aligned}$$

期待值

投影算符 $P = |x\rangle\langle x|$

$$P|\psi\rangle = |x\rangle\langle x|\psi\rangle = \langle x|\psi\rangle|x\rangle$$

- $\psi_{(n)} = \langle n | \psi \rangle \qquad \vec{v} = \sum v_i \hat{e}_i \rightarrow \vec{v}_{(i)} = \vec{v} \cdot \hat{e}_i$

$$|\psi\rangle = \sum \langle n|\psi\rangle |n\rangle = \sum |n\rangle\langle n|\psi\rangle$$

$$\sum |n\rangle\langle n| = 1 \qquad \int dn\, |n\rangle\langle n| = 1$$

$$|\psi\rangle = \int dn\, \psi(n) |n\rangle = \int dn \langle n|\psi\rangle |n\rangle = \int dn\, |n\rangle\langle n|\psi\rangle$$

- $i\hbar \frac{\partial}{\partial t} \psi(\vec{r}, t) = \hat{H}(t) \psi(\vec{r}, t)$

$$\psi(\vec{r}, t) = U(t, t_0) \psi(\vec{r}, t_0)$$

$$\psi(t) = U(t, t_0)\psi(t_0) \longrightarrow U(t_0, t_0) = 1$$

$$\psi(\vec{r}, t) = U(t, t_0)\psi(\vec{r}, t_0)$$

$$\downarrow$$

$$i\hbar \frac{\partial}{\partial t} \psi(\vec{r}, t) = \hat{H}(t)\psi(\vec{r}, t)$$

$$\downarrow$$

$$i\hbar \frac{\partial}{\partial t} U(t, t_0) = \hat{H}(t) U(t, t_0)$$

$$\downarrow$$

$$U(t, t_0) = T \exp\left(-\frac{i}{\hbar}\int_{t_0}^{t} H(t')dt'\right)$$

$$\downarrow$$

Dyson 展开

$$U(t, t_0) = 1 - i\int_{t_0}^{t} H_{int}(t')dt' + \frac{(-i)^2}{2!}\int_{t_0}^{t} dt_1 \int_{t_0}^{t} dt_2 H_{int}(dt_1) H_{int}(dt_2) + \cdots$$

$$\psi(\vec{r}, t) = \psi(\vec{r})\phi(t)$$

$$\downarrow$$

$$i\hbar \frac{\partial}{\partial t} \psi(\vec{r})\phi(t) = \hat{H}(t)\psi(\vec{r})\phi(t)$$

$$\hat{H}\psi(\vec{r}) = E\psi(\vec{r})$$

$$\downarrow$$

$$i\hbar \frac{\partial}{\partial t}\phi(t) = \hat{H}\phi(t) = E\phi(t)$$

$$\downarrow$$

$$\frac{1}{\phi(t)} \frac{d\phi(t)}{dt} = \frac{1}{i\hbar} E$$

$$\downarrow$$

$$\ln|\phi(t)| = \frac{1}{i\hbar} E t$$

$$\downarrow$$

$$\phi(t) = e^{-iEt/\hbar}$$

$$\downarrow$$

$$\psi(\vec{r}) = \sum_n C_n \psi_n(\vec{r})$$

$$\downarrow$$

$$\psi(\vec{r}, t) = \sum_n C_n \psi_n(\vec{r}) e^{-iEt/\hbar}$$

$$P_n(0) = |C_n|^2$$
$$P_n(t) = |C_n e^{-i\frac{E_n t}{\hbar}}|^2 = |C_n|^2 \; P_n(0)$$

- $\langle\psi|\hat{A}\phi\rangle = \langle\hat{A}\psi|\phi\rangle = \langle\phi|\hat{A}\psi\rangle^* = \langle\psi|A^+|\phi\rangle$

厄密算符. $A^+ = (A^*)^T$

证 $\langle\psi|\hat{x}\phi\rangle = \langle\hat{x}\psi|\phi\rangle$

$$\langle\psi|\hat{x}\phi\rangle = \int\psi^*(x)\,x\,\phi(x)\,dx = \int x\,\psi^*(x)\,\phi(x)\,dx$$
$$\langle\hat{x}\psi|\phi\rangle = \int\psi^*(x)\,x^*\,\phi(x)\,dx = \int\psi^*(x)\,x\,\phi(x)\,dx$$

证 $\langle\psi|\hat{p}\phi\rangle = \langle\hat{p}\psi|\phi\rangle$, $\hat{p} = -i\hbar\frac{\partial}{\partial x}$

$$\langle\psi|\hat{p}\phi\rangle = \int\psi^*(x)(-i\hbar)\frac{\partial}{\partial x}\phi(x)\,dx$$
$$= -i\hbar[\psi^*\phi]_{-\infty}^{\infty} + i\hbar\int\frac{\partial\psi^*}{\partial x}\phi\,dx$$
$$= i\hbar\;\frac{\partial\psi^*}{\partial x}\phi\,dx = \langle\hat{p}\psi|\phi\rangle$$

- $\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)$

$$\langle\phi|\hat{H}\psi\rangle = \int_{-\infty}^{\infty}\phi^*(x)\left(-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi(x)\right)dx$$

$$\int\phi^*\frac{d^2\psi}{dx^2}dx = \left[\phi^*\frac{d\psi}{dx}\right]_{-\infty}^{\infty} - \int\frac{d\phi^*}{dx}\frac{d\psi}{dx}dx$$
$$\downarrow$$
$$\int\phi^*\frac{d^2\psi}{dx^2}dx = -\int\frac{d\phi^*}{dx}\frac{d\psi}{dx}dx$$

$$\int\phi^*(x)V(x)\psi(x)\,dx$$
$$= \int(V(x)\phi(x))^*\psi(x)\,dx$$

- $\langle\hat{A}\rangle = \langle\psi|\hat{A}|\psi\rangle$

if $\hat{A}$ is Hermitian:

$$\langle\psi|\hat{A}|\psi\rangle = \langle\psi|\hat{A}^T|\psi\rangle = \langle A\psi|\psi\rangle = \langle\psi|\hat{A}|\psi\rangle^*$$
$$\downarrow$$

means $\langle\hat{A}\rangle$ is always real!

厄密算子的所有特征值都是实数 $\langle\hat{A}\rangle$

$$\hat{A}|\psi_1\rangle = \lambda_1|\psi_1\rangle \quad , \quad \hat{A}|\psi_2\rangle = \lambda_2|\psi_2\rangle$$

$$\langle\psi_2|\hat{A}|\psi_1\rangle = \lambda_1\langle\psi_2|\psi_1\rangle$$

$$\langle\psi_1|\hat{A}|\psi_2\rangle = \lambda_2\langle\psi_1|\psi_2\rangle$$

$\hat{A}$ is Hermitian $\longrightarrow \lambda_1\langle\psi_2|\psi_1\rangle = \lambda_2\langle\psi_1|\psi_2\rangle$

$$\downarrow$$

$$\langle\psi_2|\psi_1\rangle = 0$$

厄米算子对不同本征值的本征态正交.

- $[\hat{A},\hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$

$[\hat{A},\hat{B}] = 0$ ，两个算符对易

$[\hat{A}\hat{B},\hat{C}] = \hat{A}[\hat{B},\hat{C}] + [\hat{A},\hat{C}]\hat{B}$

$[\hat{A},\hat{B}\hat{C}] = [\hat{A},\hat{B}]\hat{C} + \hat{B}[\hat{A},\hat{C}]$

$[\hat{x},\hat{p}_x] = i\hbar$

$[\hat{p}_x,\hat{x}] = -i\hbar$

- $[\hat{x}, f(\hat{x})] = 0$ 一定!

$[\hat{x},\hat{H}] = \dfrac{i\hbar\hat{p}}{m}$

$\hat{H} = \dfrac{\hat{p}^2}{2m} + V(\hat{x})$

$$[\hat{x},\hat{H}] = \left[\hat{x},\frac{\hat{p}^2}{2m}\right] + [\hat{x},V(\hat{x})] = \frac{1}{2m}[\hat{x},\hat{p}^2]$$

$$= \frac{1}{2m}(\hat{p}[\hat{x},\hat{p}] + [\hat{x},\hat{p}]\hat{p}) = \frac{i\hbar\hat{p}}{m}$$

$[\hat{p},\hat{H}] = -i\hbar\dfrac{\partial V(\hat{x})}{\partial x} \qquad \hat{H} = \dfrac{\hat{p}^2}{2m} + V(\hat{x})$

$$[\hat{p},\hat{H}] = \left[\hat{p},\frac{\hat{p}^2}{2m}\right] + [\hat{p},V(\hat{x})] = [\hat{p},V(\hat{x})] = [\hat{p},x]\frac{\partial V(\hat{x})}{\partial x} = -i\hbar\frac{\partial V(\hat{x})}{\partial x}$$

例: $[\hat{x}^2,\hat{p}_x] = 2\hat{x}[\hat{x},\hat{p}_x] = 2i\hbar\hat{x}$

$$[\hat{x}^2,\hat{p}_x^2] = 2\hat{x}[\hat{x},\hat{p}_x^2] = 2\hat{x}\hat{p}_x[\hat{x},\hat{p}_x] = 2i\hbar\hat{x}\hat{p}_x$$

- $[\hat{A},\hat{H}] = 0 \qquad \hat{A}\hat{H} = \hat{H}\hat{A}$

$$\hat{H}|\psi_n\rangle = E_n|\psi_n\rangle \to \hat{H}(\hat{A}|\psi_n\rangle) = E_n(\hat{A}|\psi_n\rangle) \to \hat{A}|\psi_n\rangle = a_n|\psi_n\rangle$$

$|\psi_n\rangle$ 应该是 $\hat{A}$ 的本征态.

如果一个算符与哈密顿量对易, 其与哈密顿量共享一系列本征态.

- $\Delta A \Delta B \geq \frac{1}{2}\left|\langle\psi|[\hat{A},\hat{B}]|\psi\rangle\right|$

  $\langle\psi|[\hat{A},\hat{B}]|\psi\rangle = \langle[\hat{A},\hat{B}]\rangle$

  $\Rightarrow \Delta x \Delta p \geq \frac{\hbar}{2}$

- $i\hbar\frac{d}{dt}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle$

  $\downarrow$

  $\langle\hat{A}\rangle = \langle\psi(t)|\hat{A}|\psi(t)\rangle$

  $\downarrow$

  $\frac{d}{dt}\langle\hat{A}\rangle = \frac{d}{dt}\left(\langle\psi(t)|\hat{A}|\psi(t)\rangle\right)$

  $\downarrow$

  $\frac{d}{dt}\langle\hat{A}\rangle = \left(\frac{d}{dt}\langle\psi(t)|\right)\hat{A}|\psi\rangle + \langle\psi|\hat{A}\left(\frac{d}{dt}|\psi\rangle\right)$

  $\downarrow$

  $\frac{d}{dt}|\psi\rangle = -\frac{i}{\hbar}\hat{H}|\psi\rangle,\quad \frac{d}{dt}\langle\psi| = \frac{i}{\hbar}\langle\psi|\hat{H}$

  $\frac{d}{dt}\langle\hat{A}\rangle = \frac{i}{\hbar}\langle\psi|\hat{H}\hat{A}|\psi\rangle - \frac{i}{\hbar}\langle\psi|\hat{A}\hat{H}|\psi\rangle = \frac{i}{\hbar}\langle\psi|[\hat{H},\hat{A}]|\psi\rangle$

  $\frac{d}{dt}\langle\hat{A}\rangle = \frac{i}{\hbar}\langle[\hat{H},\hat{A}]\rangle$

  $\downarrow$

  $[\hat{A},\hat{H}] = 0 \longrightarrow \frac{d}{dt}\langle\hat{A}\rangle = 0$

  如一个算子$\hat{A}$与哈~$\hat{H}$对易，$\hat{A}$是一个守恒量

- 非含时薛定谔方程

  $\psi(\vec{r},t) = \psi(\vec{r})\phi(t)$

  $\downarrow$

  $i\hbar\frac{\partial}{\partial t}\psi(\vec{r})\phi(t) = \hat{H}(t)\psi(\vec{r})\phi(t) = E\psi(\vec{r})\phi(t)$

  $\downarrow$

  $\hat{H}\psi(\vec{r}) = E\psi(\vec{r})$

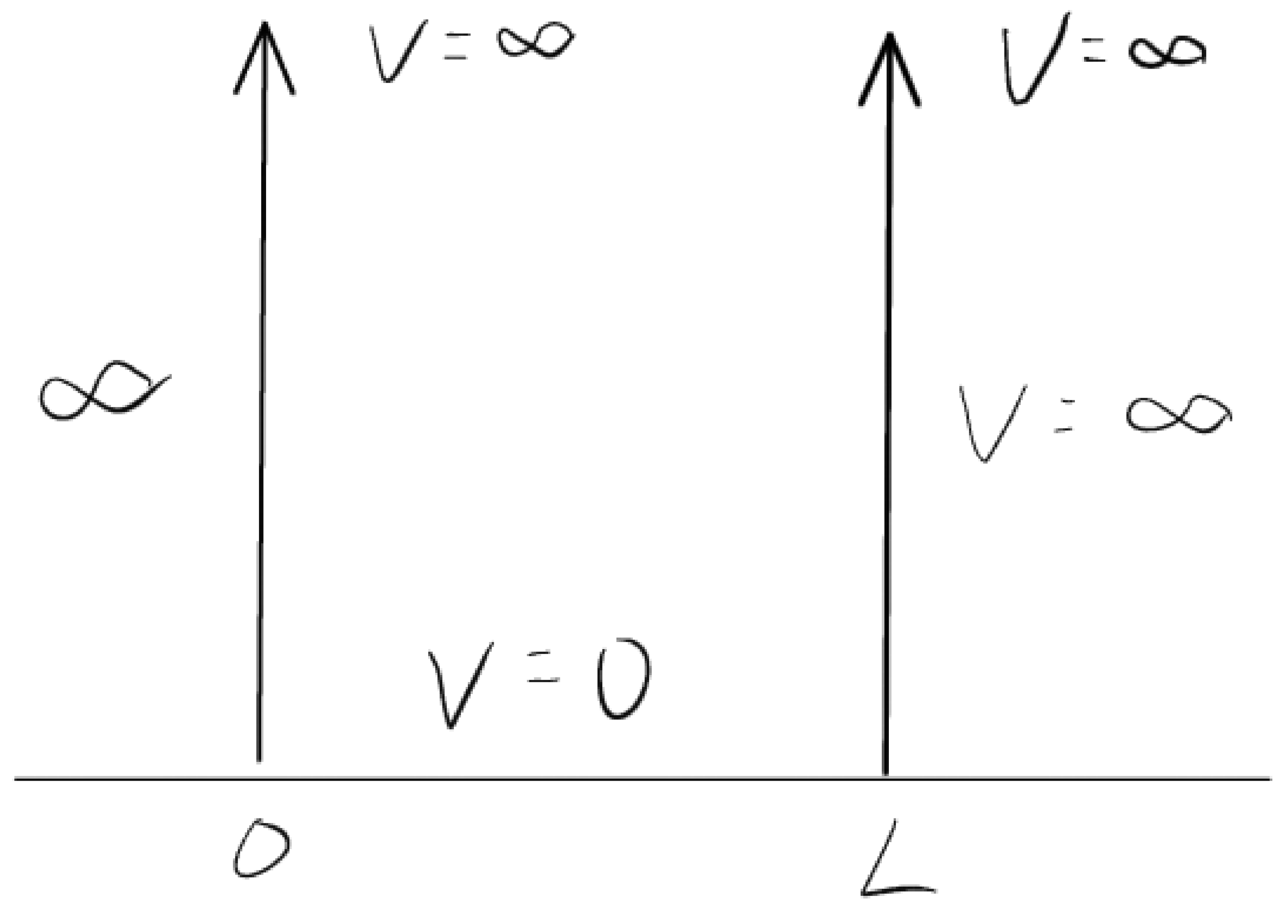
  $\downarrow$

  $\psi(\vec{r}) = \sum_n C_n\psi_n(\vec{r})$

  $\downarrow$

  $\psi(\vec{r},t) = \sum_n C_n\psi_n(\vec{r})e^{-iE_nt/\hbar}$

- 无限深势阱



$$V(x) = \begin{cases} 0 & , \quad 0 \le x \le L \\ \infty & , \quad x < 0 \text{ or } x > L \end{cases}$$

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)\right)|\psi(x)\rangle = E(x)\,\psi(x)$$

$\downarrow$

$$|\psi(x)\rangle = 0 \; : \; \text{for } V(x) = \infty$$

- For $V(x) = 0$

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\right)\psi(x) = E\,\psi(x)$$

$\downarrow$

$$\frac{d^2\psi(x)}{dx^2} = -\frac{2mE}{\hbar^2}\psi(x) \quad \longleftrightarrow \quad k = \frac{(2mE)^{1/2}}{\hbar}$$

$\downarrow$

$$\frac{d^2\psi(x)}{dx^2} = -k^2\psi(x)$$

$\downarrow$

$$\psi(x) = A\sin kx + B\cos kx$$

- Boundary condition:

$$\psi(0) = \psi(L) = 0$$

$\downarrow$

$$B = 0 \; , \quad \psi(L) = A\sin(kL) = 0$$

$$k = \frac{(2mE)^{1/2}}{\hbar}$$

for $\sin(kL) = 0$: $kL = n\pi$

$\downarrow$

$$E_n = \frac{n^2\hbar^2 k^2}{2mL} \; , \quad \psi_n(x) = A\sin\left(\frac{n\pi x}{L}\right)$$

$\downarrow$

$$\int_0^L |\psi_n(x)|^2 = 1$$

$\downarrow$

$$A = \sqrt{2/L} \; \rightarrow \; \psi_n(x) = \sqrt{2/L}\,\sin\left(\frac{n\pi x}{L}\right)$$

$\downarrow$

$$\psi_n(x,t) = \sqrt{2/L}\,\sin\left(\frac{n\pi x}{L}\right)e^{-iE_n t/\hbar}$$

- $$\langle E \rangle = \int_0^L \left(\sum_{m=1}^{\infty} C_m^* \psi_m^*(x) e^{iE_m t/\hbar}\right)\left(\sum_{n=1}^{\infty} C_n^* \psi_n^*(x) e^{iE_n t/\hbar}\right) dx$$

$\downarrow$

$$\int_0^L \psi_m^*(x)\psi_n^*(x)\,dx = \delta_{mn}$$

$\downarrow$

$$\langle E \rangle = \sum_{n=1}^{\infty} |C_n|^2 E_n$$

$$V(x) = \begin{cases} 0 & 0 \le x \le L & \text{II} \\ V_0 & 0 < x \text{ or } x > L \\ & \quad\text{I} \qquad\qquad \text{III} \end{cases}$$

$0 \le x \le L$:

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\right)\psi(x) = E\psi(x)$$

$$\Big\downarrow \; k = \frac{(2mE)^{1/2}}{\hbar}$$

$$\psi_{II}(x) = A\sin kx + B\cos kx$$

$$\downarrow$$

$$\psi_{II_n}(x) = A\sin\left(\frac{n\pi x}{L}\right)$$

Outside the well :

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} - V_0\right)\psi(x) = E\psi(x)$$

$$\Big\downarrow \; k = \frac{(2m(V_0-E))^{1/2}}{\hbar}$$

$x > L$: $\quad \psi_{III}(x) = De^{-k(x-L)}$

$x < 0$: $\quad \psi_I(x) = Fe^{kx}$

$\psi_I(x=0) = \psi_{II}(x=0) \qquad \psi_I'(x=0) = \psi_{II}'(x=0)$

$\psi_{II}(x=L) = \psi_{III}(x=L) \qquad \psi_{II}'(x=L) = \psi_{III}'(x=L)$

$$\psi_{II_n}(x) = A\sin\left(\frac{n\pi x}{L}\right)$$

$$\psi_{I_n}(x) = D_n e^{-k_n x}$$

$$\psi_{III_n}(x) = -D_n e^{-k_n(x-L)}$$

For Large V_0 : $\quad E_n \approx \dfrac{n^2\pi^2\hbar^2}{2mL^2}$

$\sum^{n} (2) = 1 \qquad 5$